Solid Geometry & Similarity

Section 1: Polyhedra & Euler's Formula

1 A convex polyhedron has 12 vertices and 30 edges. How many faces does it have?

a. 18

b. 20

c. 22

d. 24

2 What is the total surface area of a regular tetrahedron with edge length 6 cm?

a. $36\sqrt{3}$ cm^2

b. $18\sqrt{3}$ cm^2

c. 54 cm^2

d. $72\sqrt{3}$ cm^2

3 What is the length of the space diagonal of a rectangular prism with dimensions 3, 4, and 12?

a. 13

b. 15

c. $\sqrt{160}$

d. 14

Section 2: Cones, Pyramids & Frustums

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4 A right pyramid has a square base of side 10 cm and an altitude of 12 cm. What is its lateral surface area?

a. $260\ \text{cm}^2$

b. $240\ \text{cm}^2$

c. $360\ \text{cm}^2$

d. $130\ \text{cm}^2$

5 A right circular cone of radius 6 cm and height 9 cm is cut by a plane parallel to its base at height 3 cm from the base. What is the radius of the top circular cross section?

a. 4 cm

b. 3 cm

c. 4.5 cm

d. 5 cm

6 What is the volume of the frustum formed in the above cone with bases of radii 6 cm and 4 cm and height 3 cm?

a. $76\pi\ \text{cm}^3$

b. $72\pi\ \text{cm}^3$

c. $84\pi\ \text{cm}^3$

d. $96\pi\ \text{cm}^3$

Section 3: Spheres, Cylinders & Cross-Sections

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7 A cylinder is circumscribed around a sphere of radius r. What is the ratio of the volume of the sphere to the volume of the cylinder?

a. $\frac{2}{3}$

b. $\frac{1}{2}$

c. $\frac{3}{4}$

d. $\frac{4}{5}$

8 A sphere is intersected by a plane 5 cm from its center. If the radius of the cross-sectional circle is 12 cm, what is the radius of the sphere?

a. 13 cm

b. 15 cm

c. $\sqrt{119}$ cm

d. 17 cm

9 What is the total surface area of a closed hemisphere of radius 4 cm?

a. 48π cm^2

b. 32π cm^2

c. 64π cm^2

d. 16π cm^2

Section 4: Scale Factors & Volume Ratios

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10 Two similar solid statues have heights in the ratio $2:5$. If the smaller statue weighs 16 kg, what does the larger statue weigh (assuming identical material)?

a. 250 kg

b. 100 kg

c. 200 kg

d. 320 kg

11 If the volume of a sphere is multiplied by 27, by what factor does its surface area increase?

a. 9

b. 3

c. 6

d. 18

12 A solid cylinder has its radius doubled and its height halved. How does its volume change?

a. Doubled

b. Halved

c. Quadrupled

d. Remains unchanged

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